
Measuring Moral Psychology in Language Models Without Flattening It: A Theory-Decomposed Benchmark, Its Governance, and a Quarantined Computational Record

Anonymous Author(s)

Affiliation

Address

email

1 *Venue-neutral submission draft, fully anonymized, 2026-08-28. The appendix is part of this single*
2 *PDF and inherits every label on this page.*

Abstract

3 Values-alignment benchmarks routinely flatten distinct moral-psychological theo-
4 ries into one score, and blend two questions that come apart: whether a model
5 *can* reason faithfully inside a named theory when asked, and which values its *un-*
6 *prompted* behaviour expresses. This benchmark refuses both. Three schools —
7 Moral Foundations Theory, Schwartz’s refined basic-values model, and the neo-
8 Kohlbergian schema tradition — are each operationalized under their own theory-
9 native construct and never pooled. Two item families, competence and profile, are
10 separated by a mechanically enforced firewall: profile results are never ranked,
11 by construction, and never share an analysis, table or sentence with competence
12 results. The sealed instrument holds 422 public items, 280 competence and 142
13 profile, with 368 rubrics, an anti-caricature gate battery, per-item public/shadow
14 twins with an exposure ledger, and a preregistered no-result-peeking authoring
15 rule. We report a **computational** assessment over a sealed 16-model roster, the
16 families’ counts stated apart and never compared: competence covers 13 mea-
17 sured seats at 120/120 subject cells each over four never-merged substrates, 1,326
18 judged cells across 2,142 scheduled verdicts; profile covers 16 of 16 seats over
19 three pooled-eligible arcs (5 + 8 + 3). **The human expert gate (“Gate M”) that**
20 **is this instrument’s sole validity path has not run, and zero expert labels have**
21 **ever landed.** Nothing here is validated measurement or a leaderboard.

22
23 **STANDING BANNER — binds every number, table and sentence,**
24 **main text and appendix alike. PROVISIONAL · DESCRIPTIVE**
25 **· EXPLORATORY · QUARANTINED PRE-GATE-M.** The judges
26 and coders (claude-opus-5, gpt-5.6-sol) are **UNVALIDATED**;
27 Gate M — agreement against human expert labels — is the sole va-
28 lidity path and **has not run.** Every competence surface carries **PRO-**
29 **VISIONAL · pre-Gate-M,** every profile surface **QUARANTINED-**
30 **DESCRIPTIVE. Firewall G2, both directions:** competence and pro-
31 file results never blend in one analysis, table or sentence, and **profile sur-**
32 **faces are NEVER-RANKED by construction** — not pending Gate M,
33 not after it closes. Agreement between two unvalidated raters is a prop-
34 erty of the rater pair, never accuracy. Every value is transcribed from a
35 banked artifact of record; **this paper computes no new statistic.** The

36 full render law binds identically throughout and is stated in **Appendix**
37 **A**.

38 **1 Introduction**

39 Two conflations sit under most values-alignment evaluation. The first is *theory conflation*: distinct
40 traditions, each with its own object of measurement, collapsed into one alignment score. Moral
41 Foundations Theory [1, 2] asks which intuitions carry weight in context; Schwartz’s refined model
42 [3], priority ordering on a motivational circumplex; the neo-Kohlbergian tradition [4], the *structure*
43 of moral justification. Averaging them yields a number whose construct no theory recognizes — and
44 a review of 445 benchmarks reports contested or unclear construct definitions across this literature
45 [32]. The second is *question conflation*: whether a model can reason faithfully inside a named theory
46 on request, and what its unprompted behaviour expresses, license different conclusions. The first
47 admits an ordering; the second does not, because moral psychology is descriptive and none of the
48 three schools defines a *better* default profile. A benchmark that blends them exports a ranking to a
49 place where no ranking exists.

50 **What this paper reports.** A benchmark keeping three schools in their consensus forms under three
51 distinct constructs (§3.1), split across two firewalled families (§3.2); and a computational assessment
52 of a sealed 16-model roster administered hermetically, graded by model judges and coded by model
53 coders, under a census discipline treating a missing score as a re-run rather than a zero. The human
54 validation gate is designed in, blocking, and has not run.

55 **Contributions.** (i) All three schools operationalized together, each under its own theory-native con-
56 struct, never flattened into one score (§3.1). (ii) The never-ranked firewall as an *enforced reporting*
57 *rule* — not the competence/disposition distinction, which is prior art (§3.2). (iii) A quarantined com-
58 putational record, a portable administration protocol, and the failure modes the architecture made
59 visible (§5-§8). Per-item twins with an exposure ledger, the no-result-peeking rule, and sealed pools
60 with an amendment ledger are offered as documented practice rather than priority (§3.3-§3.5).

61 **2 Related work and position**

62 **Instruments on models, and multi-theory work.** A mature line, and we claim nothing in it:
63 Moral Foundations benchmarks exist [17], and Defining Issues Test administrations report post-
64 conventional scores [18] substantially inferior in some of six tested languages, consistent with form-
65 matching rather than principled reasoning [19, 43]. Every pairwise subset of our three schools
66 is occupied — a nine-questionnaire compilation of 600+ items administers a Moral Foundations
67 Questionnaire-2 and the Schwartz Values Survey together but no Defining-Issues-family instru-
68 ment [14]; an ACL 2026 paper makes Schwartz’s refined 19-value circumplex the *primary coordi-*
69 *nate system* and argues against flat scoring [15]; Kohlberg-stage scoring of model *free text* by
70 a validated judge pipeline spans 13 models [16]; others use both lenses on escalating dilemmas,
71 Schwartz-grounded scaffolds, or agent-scale value evaluation [42, 46, 48]; a benchmark titled for
72 three-dimensional moral evaluation varies metrics, not schools [49]. To our knowledge, no exist-
73 ing benchmark measures all three schools — Moral Foundations, Schwartz, and neo-Kohlbergian
74 reasoning — each under its own theory-native construct and never flattened into a single score.
75 We claim that three-way conjunction and nothing weaker: not first to preserve a theory’s native
76 structure [15], not first to score Kohlbergian reasoning in free text [16], not first to combine two
77 moral-psychology instruments in one benchmark [14]. A position paper argues that descriptive
78 frameworks should *not* proxy for norm-level evaluation [11]; we accept that scope limit rather than
79 contest it. **Appendix F sets the nearest systems beside this one, row by row.**

80 **The two-family split.** The distinction is prior art: a Nature roadmap owns moral performance versus
81 moral competence [10], the position paper above the value-problem/norm-problem split [11], an
82 ACL benchmark evaluates value *orientation* alongside value *understanding* over 44 inventories [12],
83 others separate declared from enacted values and deep from shallow generalization [41, 45]. Most
84 directly, a Schwartz-grounded platform reports discriminative and generative-conformity evaluation
85 separately, shows they diverge across 33 models — and *ranks* models on the conformity score [13].
86 Neither the split nor the divergence is ours. What we operationalize is a rule: the profile family is

never ranked, tiered or composited, and never appears in the same table or sentence as a competence result, in either direction.

Contamination. Every ingredient is occupied: public-plus-withheld-private splits [26], scheduled refresh with a permanently withheld fraction [27], contamination-resistance-by-construction as an accepted position [28], psychometric-instrument contamination across 21 models including a Schwartz-family inventory [24]. What we did not find in a values benchmark is *per-item paired twins plus an exposure ledger*, and that is the narrow claim. Its limit: our n-gram gate is verbatim-leak hygiene, and semantic duplicates lift performance on genuinely held-out items by 13-19 percentage points (ecological) to 22 (controlled) [25].

Rubric authoring. Expert-written criteria with a human-validated judge are published — 1,000 scenarios plus a 150-example theory set, 23,018 criteria, a judge validated against two experts at Cohen’s kappa 0.75 [20] — but without documented blinding to model outputs, so our claim is about *documented discipline*, not solitude: our rule forbids any model call in an authoring session, the opposite of the norm in which annotators craft items against observed failures [21]. A counter-current is live — model-generated rubrics can be better calibrated to human rubrics than model responses are [23] — read here as calibration to an existing rubric, not fidelity to a theory, against the reported human/model rubric gap [22].

Preregistration, governance and denominators. No priority claim here. Preregistration for AI evaluation is a venue-accepted 2026 program, one mechanism’s authors finding 6 of 7 previously p-hacked configurations failed to transfer to a new model [29, 30]; governance is occupied by a lifecycle framework [31], the 445-benchmark review above — only 16.0% of its benchmarks run any statistical testing [32] — a reporting schema [33] and a safety-domain standard [47]. Our sealed pools, hashed manifests and amendment ledger are a domain-scoped *instantiation*; we are not aware of another preregistered values or moral-psychology benchmark — what we did not find, never a claim of priority. Denominators likewise: coverage density as a headline is HELM’s founding contribution (17.9% to 96.0%) [35], denominator dishonesty is documented [36], item-level releases ship as infrastructure [34] — but that schema has no field for refusals, failed responses or excluded items, a gap we verified rather than inferred, so we report a coverage arithmetic naming every non-measured cell and its reason, a practice worth adopting rather than a first.

Judge validity. The model-as-rater paradigm arrived with its agreement caveats attached [9]; validating judges against humans is now a norm, with a formal substitution test [38], a position paper arguing judges are neither demonstrably valid nor reliable [37], and evidence that raw agreement overstates judge quality — kappa deflation of 33-41 points on one benchmark across 21 judges [40]. Rare is compliance: fewer than 8% of such papers also run human validation [39]. Our quarantine is practice, not invention, and we sit behind the moral-domain state of the art, where validated moral-reasoning judges already exist [20, 16]. §8 says why the gate is undischarged; the answer is architectural, not budgetary.

3 Benchmark design

3.1 Three schools, three constructs

Three schools are measured under deliberately distinct constructs: Moral Foundations Theory (intuition weighting), Schwartz basic values (value-priority structure), and neo-Kohlbergian reasoning (reasoning-structure sophistication). Scores are never aggregated across schools; there is no composite moral-psychology score. The constructs reach into the coding schemes, which Appendix A states in full; three properties are load-bearing here. The Moral Foundations construct is an intuition-weight profile over six foundations in sealed instrument order, with Liberty never scored — a design rule, not an omission. The Schwartz construct is circumplex-gated: a *structure gate on raw ratings*, whose unit is a pseudo-respondent bundle with a sealed minimum of 30 per seat, must pass before any priority vector may be reported; the gate unit is the seat, seats never pooled. The neo-Kohlbergian construct is a schema distribution in a sealed declared order **that is not a scale**; stage labels and developmental language are never used and no human-norm comparison is permitted. Both exclusions are design law, not defect.

3.2 Two families and the firewall

Family C and Family P are disjoint pools with disjoint scorers and disjoint report sections; the render layer has no code path joining them. Family C is ranked-capable and is what this paper's competence results describe. Family P is descriptive: its licensed claim form is that a model's default behaviour expresses a given weighting, priority or structure with a given dispersion under stated conditions, forbidding any better/worse statement, ranking, trait language, or "model X has values V" simpliciter. Never-ranked is structural rather than provisional, which is why no Family-P board exists at any stage. The difference runs to the raters: Family C uses a **judge**, producing a graded score that can feed an ordering; Family P a **coder**, producing a profile code object — never a grade, never reduced to one code, never ordered. Two independent codes sit side by side because, absent gold labels, disagreement is the datum.

3.3 Item and rubric law: anti-caricature gates and no result-peeking

Authoring order is fixed and preregistered: construct specification and the school's verification rubrics first, then a commentary entry recording construct target, difficulty rationale and target-pair note, and only then the public item, its rubric, and its shadow twin. That entry is also the anti-p-hacking mechanism, recording before authoring which models and behaviours the item is expected to separate and why, **from construct difficulty alone, never from observed outputs**. The governing rule is law: "*Never look at any model's output on any drafted item. Authoring sessions do not call models, full stop.*" A companion rule keeps audits independent — a session never audits its own pool. A validator battery enforces theory fidelity mechanically; its most load-bearing member is an anti-cueing gate forbidding competence item surfaces from naming any school's theory terms, value labels, foundation names or schema names, so only rubrics and coding instructions speak theory. Appendix C lists the battery at class grain; Appendix G illustrates the item shape.

3.4 Twins, exposure, and the limits of the contamination design

Every public item has exactly one shadow twin, matched on school, setting, difficulty, adversity and construct targets, with n-gram overlap checked across all text surfaces. Both halves are authored by the same session from the same construct note with surfaces *independently generated* — not edit-distance paraphrases, which the n-gram gate exists to prevent. Custody is asymmetric: the shadow pool is excluded from the released record from the outset, and the seal pins shadow *hashes*, not text.

All 280 Family-C items carry a shadow identifier naming a never-released paraphrase twin (280/280). That is the only shadow count this paper carries; no instrument-wide shadow-twin total is asserted anywhere. The prospective contamination indicator is the public-shadow score gap per model per pool against a temperature-zero baseline: about zero at release, a growing public-side advantage in re-runs is the alarm. Exposure is ledgered rather than assumed, and the ledger says the pre-declared release precondition is **not met instrument-wide** (Appendix D). A separate screen forbids 8-gram overlap with canonical instrument strings held locally; the Defining Issues Test is proprietary and not held locally, so its screening is structural rather than lexical — a standing disclosure.

3.5 Sealed preregistration and amendment governance

After the design freeze the sealed directory, the sealed constants, and the pool, rubric and shadow hashes are law, enforced by a signature lock and a pool-drift guard. **A change to a sealed constant, pool or rubric requires a ledger entry recording old value, new value and reason, plus a seal rotation — never a silent edit**; entries that only read sealed state, or record a status, rotate nothing, and the ledger says so. The seal script seals on *any* invocation and is therefore never probed: it ran once inside ratified amendment A-14 — the re-seal of record — and once inside A-21. The register's headings run A-1 through A-22 (Appendix E, header grain); one draft, A-3, was withdrawn and never existed as a ratified amendment. Two entries are load-bearing here. A-1 let a *provisional exploratory* board precede Gate M provided it carries mandatory labelling and render rules, and quarantined every cell it covers "until re-scored under a Gate-M-validated panel or Gate M validates the panel as configured" — both disjuncts always travelling together, with a standing caveat that A-1 labelling and render rules are applied while full conformance to its ranking clause is not claimed. A-17 replaced a substrate-join rule with a two-stratum disclosure after a forensic pass showed a 9,280-cell arm *labelled* at one sampling temperature and *administered* at another.

190 4 Administration, judging, and census

191 **Hermetic administration, probing, and custody.** Three routes are used, only the metered API
192 route cash-bearing. Hermeticity is a measurement control: a contaminated tool-configuration direc-
193 tory loads unrelated skills into a subject or coder call and yields slow or empty responses. Concur-
194 rency is pinned, not chosen — a ceiling of 8 on one judge/coder route, a hard 1 on the other — and
195 the assertion never clamps, it refuses. Any action premised on a carried belief about external state
196 requires a fresh probe within 24 hours, because *reachability is a parameterised fact*: four seats were
197 “served and charged at $t = 0.0$ and refused at $t = 0.7$, ninety minutes apart, on the same key and
198 the same host.” Custody follows from a forensic finding: a scan of one 14,122-file run tree found
199 no outgoing payload or parameter echo anywhere, so every stored parameter was a harness-written
200 label. An administration now banks a per-cell digest of its outgoing payload **before** transport, and
201 close-out claims are verified **by checksum**.

202 **Judging and coding legs.** The competence family applies per-subject leave-one-out in-family judge
203 exclusion, which is why some substrates are dual out-of-family and others single, at 204 and 102
204 verdicts per seat over the same 102 judged cells — a firewall consequence, not a coverage deficiency.
205 The profile family deliberately has **no** leave-one-out, there being no rank to corrupt, so the remedy
206 is disclosure: 7 of the 16 seats are in-family for exactly one coder leg — 3 against leg A, 4 against
207 leg B, none for both. The two legs code the same cells, are **never pooled for agreement**, and run
208 sequentially with caches keyed on content **and** rater parameters.

209 **Blinding, census, and cost.** Rater prompts are swept at build time for judge and subject slugs, ven-
210 dor names and path fragments, in every leg and the dry run (Appendix A). Census-clean means
211 zero *unexpected* nulls with structural nulls pre-declared: read each cell’s score field, never file
212 counts or exit codes; a null means re-run, never zero, never impute, never drop; denominators are
213 *scheduled* cells. One predicate defect is disclosed rather than papered over — the judge-leg flag is
214 completeness-blind, and one banked leg read clean at 206 of 210 records with an abort flag set —
215 so **every census flag travels with its N/N count**. Cost is disclosed as process, never as a result, and
216 **no program-total reconciled cash figure exists in either family** (Appendix A).

217 5 The computational assessment

218 *Two separate reports follow. Under Firewall G2 they share an architecture and*
219 *nothing else: no count, statistic or sentence below crosses between them.*

220 **Why print a board at all, under quarantine?** Because the alternative is the failure this architecture
221 exists to prevent. Exclusions, retractions and re-labels are *rendered* rather than deleted, so a reader
222 sees which seats were declared absent and why; the per-seat record is the evidence for §6’s findings;
223 and withholding it while printing the aggregate would be the denominator dishonesty §2 objects to
224 in others. Quarantine forbids interpretation, not publication.

225 5.1 Family C — competence, PROVISIONAL and pre-Gate-M

226 **What was administered, and what was declared absent.** Of the 16 sealed roster seats, **13 are mea-**
227 **sured, each at 120/120 subject cells**, across four never-merged substrates (Appendix B: a coverage
228 map, never a board). The landed reasons-route surface is **1,326 judged cells over 2,142 scheduled**
229 **verdicts** ($306 \cdot 612 \cdot 204 \cdot 1,020$ by substrate) — a matrix total, never a board. Those 13 are **the**
230 **board of record**: 16 sealed seats minus three declared absences, each with a reason. gpt-5.5 is a
231 transport failure declared after two bounded attempts and renders as an **EXCLUDED row at denom-**
232 **inator 0** carrying no result; transport was later proven by a neutral probe, but 3/3 smoke episodes
233 were prose-only and the declaration is terminal absent a queued amendment. openai/gpt-oss-20b
234 and google/gemma-3n-e4b-it are excluded as never-scheduled, contributing 0 to every board cell.
235 “Absent” means absent from the board of record, not never measured anywhere.

236 **Census, and the board.** Every administered substrate is census-clean on its own declared denomi-
237 nators; where a flag reads false it does so **solely** by way of a declared roster reduction, so a shortfall
238 renders as a declared exclusion rather than a smaller denominator, and the board’s single no-verdict
239 is a judge safeguard refusal excluded from its stratum’s denominator and never imputed (Table B1).
240 Per-seat means over each school-by-setting stratum are banked at 189 board cells across 21 seat-

rows, rendered in Appendix B one table per substrate and per judge, in census or source order, never sorted by any value, each carrying its pre-observation stratum-size labels. The only banked grand means — the five-seat dual-judged substrate’s — are there too, per leg and **never averaged into one number**; grand means for the other three substrates are **not published or banked anywhere**.

Instrument-quality results, about the instrument and never about a seat. A pre-registered fidelity sample returned STOP; the commissioned census found most audited expansion pairs defective, dominated by a class in which rubric criterion text inherits the public half’s surface idiom. Its direction is a standing statement: every coupling gave the judge literal string matches on the public half while demanding semantic mapping on the twin, **attenuating the public-shadow gap, i.e. biasing the contamination indicator toward “no contamination.”** Appendix D carries the rate, interval, strata, repair effect and the two never-pooled twin-baseline epochs.

5.2 Family P — profile, QUARANTINED-DESCRIPTIVE and never ranked

QUARANTINED-DESCRIPTIVE · NEVER-RANKED — absolute, by construction (Firewall G2, both directions). Coder legs are never collapsed, substrates never merged, the two value-priority strata never placed side by side.

Coverage, and a rendering decision. The coded surface covers **16 of 16 seats** over three pooled-eligible arcs (5 + 8 + 3), each census-clean on its own leg denominators; a fourth, out-of-pool arc replicates the same five seats and is cited for **ordering replication only, never pooled**. The per-seat cell frame and the pooled coded denominators are in Appendices A and C, endorsement-inclusive and travelling with a containment statement rather than a revised denominator. **204 of every seat’s 274 profile cells — 74% — never reach a coder at all**, the Schwartz channel being mechanically extracted. This paper prints no per-seat profile quantity: those shares are banked at both legs, but rendering them side by side invites the ordering the firewall forbids, so we report at channel, foundation and condition grain and state the omission.

The value-priority construct: absent by construction, then stratified. At temperature zero a repeat administration of the same variant is a byte-identical draw, so each seat yields exactly 3 bundles against a sealed floor of 30 and the preregistered DECLINE fires structurally — non-recovery being a property of the condition, **never a model disposition**. Two dedicated arms then administered the construct at the floor; they are **two strata, never one surface**, never compared, pooled or differenced, and are rendered in separated provenance-first blocks in Appendix C. One reading binds both: **absence of structure may not be claimed for any seat**.

Coverage arithmetic, in force. A coverage figure is reported with its strata, or not reported. The arithmetic of record is $(7+5)/16 = 12/16$ **sealed-regime + the fully-retired temperature-zero stratum (0 residual rows) + 4 provider-impossible**. The +5 are five seats *re-administered* census-clean at the sealed regime and admitted per seat under amendment A-19 — new administrations inside that stratum, **not** the temperature-zero stratum carried across the fence. A mandatory disclaimer travels with the figure: an earlier “12/16” was a **ruled-false** claim about a *different* state, and every annotated surface carries the correction “Any ‘12/16 under the sealed regime’ figure herein is RULED FALSE.” The four provider-impossible seats are transport facts about a parameter, with **no cell administered** (Appendix C).

Agreement, which is not accuracy. Field-wise exact agreement between the coder legs, by arc and never pooled, and Cohen’s kappa [7] — emitted for the six foundations and the Kohlberg primary field only, the other three channels having **no chance-corrected counterpart** — are in Appendix C.

6 What the assessment shows

Every statement below is a bounded instrument, judge-pair, coder-pair or condition property, hedged as its source hedges it; none is a validity claim or a seat verdict. The families are reported in separate subsections and no sentence joins them.

What we assert. Five claims, each tagged with its family as §8 tags its limitations: (1) [C] *the judge-pair offset direction replicates* — on two independent dual-judged substrates the signed mean delta has the same sign, a rater-pair property under one retired prompt regime, no magnitude claim; (2) [C] *high agreement can be cheap* — the closest-agreeing stratum on the larger substrate is a

292 FLOOR stratum, so agreement magnitude is confounded with stratum position; (3) [C] *the rubric-*
293 *echo class biases the contamination indicator toward “no contamination”* — a direction, not a size,
294 reaching the administered pool; (4) [both] *a label is not an administration*, three receipts attesting
295 an assertion rather than the act; (5) [P] *the coder-channel agreement ordering replicates* on all
296 three pooled-eligible arcs and, as an ordering, out of pool. None crosses the firewall, and none is a
297 validated measurement.

298 6.1 Competence-side findings

299 *PROVISIONAL · pre-Gate-M. Judges UNVALIDATED; every reading below is a*
300 *descriptive reading by an unvalidated judge, not a validity claim.*

301 **Headroom, scoped.** Within its own substrate and denominator, the Anthropic band (substrate B1)
302 has an agentic stratum sitting high (.80-.98) without being a point mass and a multi-turn stratum
303 that spreads wide — Schwartz multi-turn spans .386-.891 across the three seats, Moral Foundations
304 multi-turn the band .66-.82 its source prints. **Scoped to those two settings:** several of Table B3’s
305 single-turn cells sit at or adjacent to CEILING, a design input never read as a seat verdict.

306 **The offset, and why agreement is not accuracy.** The two dual-judged substrates are reported
307 separately, denominators named, never pooled (Table B10). The signed mean delta is **+0.0403**
308 on 306 paired cells and **+0.0599 [OPEN ESCALATION]** on 510 — the second carries that label
309 wherever quoted — so gpt-5.6-sol reads higher on average than claude-opus-5 on both, and
310 the *direction* is the only licensed cross-substrate statement. The closest-agreeing stratum on the
311 larger substrate sits at mean absolute delta **0.030** on $n = 70$, band label FLOOR: agreeing that a cell
312 is at the bottom of the scale is cheap.

313 **Coverage is not performance.** The board of record is 13 declared seats over 120 of 280 items —
314 42.9% coverage, **0 of 160** expansion items measured anywhere — graded by two unvalidated judges
315 under a retired prompt regime, in a rubric epoch that is not the current one.

316 6.2 Profile-side findings

317 *QUARANTINED-DESCRIPTIVE · NEVER-RANKED. Coders UNVALIDATED;*
318 *every reading below is a channel, condition or coder-pair property, never a seat*
319 *property.*

320 **Channel effects replicate.** Field-wise exact agreement between the coder legs orders the kohlberg
321 coding fields **register-without-structure > primary > secondary > confidence**; that ordering holds
322 on all three pooled-eligible arcs and replicates out of pool (Table C2) — a property of the channels,
323 never of a model.

324 **High agreement can be a base-rate artifact.** Purity separates at **0.943** pooled while the other five
325 foundations sit in a **0.63-0.83** band whose internal order does not replicate across arcs, so the honest
326 Moral Foundations statement is *one separation, not a ranking of six*. Purity also carries the largest
327 kappa on every arc while carrying the smallest salient share on every seat: partly a base-rate effect
328 [44], but not *only* one (Table C4).

329 **The central methodological finding is a governance one.** All 9,280 cells of a banked arm were
330 administered at temperature 0.0 under a temperature 0.7 label while **three** receipts appeared to assure
331 the opposite; the disposition was a public re-label, not a quiet re-render. Response statistics could
332 not adjudicate: two runs that provably send temperature 0 agree byte-for-byte on only **0.0-12.6%** of
333 same-prompt pairs, so the discriminant has no power — verdict INDISTINGUISHABLE, its weak
334 lean disclosed as a lean.

335 **A whole school is missing.** No validated per-seat 19-value priority profile exists anywhere: **the**
336 **Family-P profile record covers two schools of three.**

337 7 Using the benchmark on new models

338 The protocol is portable and is the part of this work usable today — a ladder whose every rung is
339 refusable.

340 **1. Probe at the parameter block the run will send, then administer hermetically.** Probe identity
 341 and reachability inside 24 hours, from inside the invoker and on the calling thread, sending the
 342 *exact* administered parameter block; a preflight omitting a parameter cannot see an impossibility
 343 at it. Strip the environment, bank a digest of each outgoing body *before* transport, and interleave
 344 administrations so a mid-run halt leaves a balanced partial design.

345 **2. Rate under the family's own rule.** For competence, apply per-subject leave-one-out in-family
 346 exclusion and accept its price: single out-of-family bands carry no agreement surface, because in-
 347 family grading is a coupling no agreement statistic can detect. For profile, do not — there is no rank
 348 to corrupt — but enumerate in-family seats by name. Run legs sequentially; never collapse them
 349 into one code.

350 **3. Take a census, then report coverage as an arithmetic rather than a fraction.** A null score
 351 is a re-run target, never a zero or a silent drop; the denominator is the *scheduled* cells; every flag
 352 prints with its N/N count. An unmeasurable subject is declared terminal with a named mechanism
 353 and stays in the denominator; an unadministrable seat renders at **denominator zero** with a reason.
 354 Where one number would require pooling across a fence, render the strata instead, distinguishing
 355 declines by design, permitted-but-failed and terminal-unmeasured.

356 **4. Claim only what the pre-validation posture licenses.** A user may report per-seat competence
 357 readings labelled provisional, with rater identity, denominator and stratum-size label; descriptive
 358 profiles in each school's own vocabulary; and rater agreement *as a rater-pair property*. A user may
 359 **not** report a ranking, tier, composite or best/worst statement of any measured seat; may **not** rank
 360 profiles at all, then or ever; may **not** treat agreement as accuracy; may **not** merge substrates, average
 361 coded legs, average foundation shares into one number, treat the three kohlberg schemas as a scale,
 362 or place two strata side by side. A re-run after a rater-regime change is a **new measurement**.

363 8 Limitations

364 Tagged [both] architecture-wide, [C] competence, [P] profile.

365 [both] *Computational assessment; the human gate has not run.* Gate M — human-expert valida-
 366 tion of the judge and coder panels — is the sole validity unlock, and **zero expert labels have ever**
 367 **landed**, mechanically re-verified; every number here is an unvalidated rater's output or an agreement
 368 statistic between two. The gate is not unfunded but *designed in and blocking*, and under-specified
 369 in a way funding does not fix: **no numeric minimum detectable effect exists**, its vehicle unrul-
 370 ed, its recruitment track open. It is **school-asymmetric by sealed design**, allocating cells to two of
 371 the three schools and zero to the third. A judge-substitution test on an annotated subset [38] is no
 372 substitute: the gate's unit is agreement with *psychometrically trained, school-native* coders whose
 373 judgement is the measurement [5, 6], and raw judge-human agreement cannot discharge it where
 374 chance correction deflates agreement by tens of points [40]. *Nothing is expert-validated, confir-*
 375 *matory, robust-panel, stable-ordering, finalized or submission-ready.* *Single administration:* every
 376 board cell is one administration, profile dispersion is ruled none, neither family carries a variance
 377 estimate. *Extent exceeds the evidence base:* 160 competence items never-administered and blocked;
 378 16 shadow surfaces at zero exposure. *Contamination is bounded only for verbatim leakage* [25].
 379 *Scope:* no ordering of models, no confirmatory analysis, no hypothesis verdict, no generalisation
 380 beyond the measured seats. A later single-seat expansion and a variant-dispersion analysis ran after
 381 this record was fixed and are **not incorporated**.

382 [C] *The landed grades are not individually auditable:* no evidence quotes were stored, no judge-
 383 template hash sealed. A later amendment repaired the template forward — labels-plus-evidence, a
 384 verbatim quote required on every gate and non-pass grade — but two amendments retired all prior
 385 judge caches by design, and pre- and post-repair rubrics are **different instrument epochs: never**
 386 **pooled, averaged or differenced.** *The rubric-echo class reaches the administered pool:* of 47 open
 387 echo carve-outs, **42 sit in the pre-expansion pool and 5 in the later wave**, and the pre-expansion
 388 pool is the administered board — the repaired 144-pair pool is clean of the class, the administered
 389 120 are not, and the standing non-independence rule requires any twin-gap analysis over it to treat
 390 rubric echo as an undisposed attenuation-direction bias. A further open escalation is a defect class
 391 found systematic at **6 confirmed defects in 18 spot-checked pairs (33%, Clopper-Pearson 95%**
 392 **CI 13.3-59.0%) plus 6 uncounted uncials**, beyond the census estimate and the repair's ruled
 393 scope; **no pool-wide rate exists and none may be constructed.** Two judging-path defects are

open: a 0-based/1-based mismatch points judge prompts one turn early on one school's multi-turn rubrics, and criterion identifiers carry surface-coupled lexemes rendered to judges. *Substrates are not comparable* — different seats, dates, routes, judge stacks and denominators — and the same 120 items reached the same 5 seats on two substrates, so an independence caveat rides any cross-substrate statement.

[P] *Never ranked, structurally*, as §3.2 states. *Reliability describes a minority of the surface* — every agreement statistic is on the 70 coded cells per seat, 26% of each seat's 274 — and *chance correction is partial*, kappa existing for the six foundations and one kohlberg field only. *Coder independence is partial by disclosure* on 7 of 16 seats, contained by the never-rank law rather than solved by it. *One construct is stratified with no gate lift path as built*: the value-priority channel is mechanically extracted with no coder, so Gate M cannot reach it and the Family-P profile surface reports two schools of three; its *two strata may never be compared* across the temperature fence. *Variant dispersion is unmeasured on the coded arcs*. *A stated-endorsement layer carries an adverse fidelity finding* — **8 of 16 reviewed pairs confirmed defective (50.0%, Clopper-Pearson 95% CI 24.7-75.3%)** in two schools, **3 of 8 (37.5%, CI 8.5-75.5%)** in the third, with a **directional** exclusion, overlapping intervals, no cross-arm rate claim, and reviewers themselves unvalidated (Appendix C). *Two rulings bar surfaces outright*: no scored consumption of one coding channel until an administered path consumes its repaired rubric, and no per-cell public-versus-twin comparison on the 76 single-variant profile items until an order-synchronization design exists. *Three internal baselines were specified and never computed* — a uniform null, a social-desirability null and a verdict-justification coherence check, named as future work rather than quietly dropped; the social-desirability annotation on all 142 items feeds no analysis code; and *one arc banks no pool snapshot*, so it is excluded from re-extraction rather than estimated.

[both] **Language and cultural scope**. The sealed language scope is one condition — English models administered in English — so nothing here speaks to behaviour in another language. The three schools' norms derive largely from Western, educated, industrialized populations — the motivating problem of the six-foundation instrument we adopt [2] — and the cross-language result we cite reports substantially inferior post-conventional scores in some languages on one battery [18, 19]. **Cross-cultural and cross-linguistic invariance is unmeasured here.**

9 Conclusion

A values benchmark can decline to flatten its theories and to blend its questions, at a legible cost: separate constructs, pools, raters and reports, and a coverage arithmetic longer than a fraction. What that buys is the ability to say precisely what was *not* measured — a school absent by construction, four seats impossible at a parameter, 160 items never administered, one stratum whose label never described its own administration — without imputing any of it. The record is a quarantined computational assessment whose sole validity path is a human expert gate that has not run, and we report it in that posture because the alternative — publishing model-judged numbers as findings while the validation channel stays unbuilt — is what this architecture exists to refuse.

10 Reproducibility statement

Reproducibility of the *computation* over stored records is a different claim from reproducibility of the *grading*, and only the first holds: every judge cache produced before two rater-regime amendments was retired by design, so **repeats are new measurements, never reproductions**. Sealed pools, rubric hashes and manifest digests fix the instrument; sealed analysis constants fix the bootstrap unit, resample count and seed; census, custody and blinding gates are tested code. The competence bundle ships as a versioned, pre-Gate-M release carrying quarantine labels on every results surface — 120 board-measured items of 280, post-repair rubrics, seal manifests and engine modules, under CC-BY-4.0 for items, rubrics and documents and MIT for code — verified by a fresh copy checked against its manifest of record at **exactly 29 files, all 28 manifest rows byte-exact, 0 mismatches**.

What a third party can and cannot verify. *Can*: that the released competence items and rubrics are the sealed artifacts byte for byte; that the geometry, gates and constants are as described; that the arithmetic over released surfaces reproduces; that the printed labels are the labels the artifacts carry.

446 *Cannot*: the grading, because its caches were retired by design and a re-grade is a new measurement;
 447 the never-administered expansion, withheld to preserve its clean pre-exposure era; and **the entire**
 448 **profile family, excluded from every released surface by custody law**. That exclusion is an **artifact**
 449 **limitation of this work**: the Family-P record of §5.2 and Appendix C is not independently checkable
 450 at item grain today, so its coverage, agreement and stratum figures rest on disclosure discipline rather
 451 than inspection. Publication is not validation.

452 11 Ethics and broader impact

453 This instrument measures something people reasonably worry about being measured: what moral
 454 considerations a model’s unprompted behaviour expresses. Our central commitment is that such
 455 measurement **must not become a ranking**. Moral psychology is descriptive, no school here de-
 456 fines a better default profile, and a benchmark emitting one would launder a contested normative
 457 judgement through an apparatus that looks empirical. The profile family is therefore never ranked,
 458 tiered or composited, by construction rather than policy — **and we decline to publish a moral**
 459 **leaderboard although the machinery to compute one exists in our own pipeline**. The compe-
 460 tence family admits an ordering, but measures reasoning *within a named theory on request*, not
 461 moral quality; reading a competence score as moral endorsement is the category error the firewall
 462 prevents.

463 The dual-use risk is real: a per-model value profile could select models for value-congruent per-
 464 suasion, or certify one as morally aligned on unvalidated ratings. Both misuses need a ranking or
 465 a validity claim; this paper supplies neither, and the released bundle excludes the profile family
 466 entirely.

467 No human-subjects data has been collected. Gate M is designed as an expert-panel validation with
 468 a sealed cell packet and sits at the design stage: recruitment is open, **zero expert labels have ever**
 469 **landed**, and this paper reports no human judgement of any kind. Any future panel is human-subjects
 470 work under its own review, consent and compensation terms; nothing here pre-authorizes it.

471 References

472 (*Markers [1]-[49] mark concept and prior-work provenance. No displayed value describing this benchmark*
 473 *derives from them; prior-work figures quoted in §2 are the cited works’ own.*)

- 474 1. J. Graham, J. Haidt, and B. A. Nosek. Liberals and conservatives rely on different sets of moral
 475 foundations. *Journal of Personality and Social Psychology*, 96(5):1029-1046, 2009.
- 476 2. M. Atari, J. Haidt, J. Graham, S. Koleva, S. T. Stevens, and M. Dehghani. Morality beyond the
 477 WEIRD: How the nomological network of morality varies across cultures. *Journal of Personality*
 478 *and Social Psychology*, 125(5):1157-1188, 2023. (MFQ-2, the six-foundation form used here.)
- 479 3. S. H. Schwartz, J. Cieciuch, M. Vecchione, E. Davidov, R. Fischer, C. Beierlein, A. Ramos, M.
 480 Verkasalo, J.-E. Lonnqvist, K. Demirutku, O. Dirilen-Gumus, and M. Konty. Refining the theory of
 481 basic individual values. *Journal of Personality and Social Psychology*, 103(4):663-688, 2012.
- 482 4. J. R. Rest, D. Narvaez, S. J. Thoma, and M. J. Bebeau. DIT2: Devising and testing a revised instru-
 483 ment of moral judgment. *Journal of Educational Psychology*, 91(4):644-659, 1999.
- 484 5. L. J. Cronbach and P. E. Meehl. Construct validity in psychological tests. *Psychological Bulletin*,
 485 52(4):281-302, 1955.
- 486 6. S. Messick. Validity of psychological assessment. *American Psychologist*, 50(9):741-749, 1995.
- 487 7. J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measure-*
 488 *ment*, 20(1):37-46, 1960.
- 489 8. C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the
 490 binomial. *Biometrika*, 26(4):404-413, 1934.
- 491 9. L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H.
 492 Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena.
 493 *Advances in NeurIPS 36* (Datasets and Benchmarks Track), 46595-46623, 2023.
- 494 10. J. Haas, S. Bridgers, A. Manzini, B. Henke, J. May, S. Levine, L. Weidinger, M. Shanahan, K. Lum, I.
 495 Gabriel, and W. Isaac. A roadmap for evaluating moral competence in large language models. *Nature*,
 496 650(8102):565-573, 2026.
- 497 11. E. Kierans, A. Dutt, J. Rittichier, S. Dori-Hacohen, and A. Ghosh. Position: Evaluations of AI moral
 498 reasoning still miss half of the picture. ACL 2026 Workshop on Evaluating Evaluations (EvalEval),
 499 archival; arXiv:2608.14566.

12. Y. Ren, H. Ye, H. Fang, X. Zhang, and G. Song. ValueBench: Towards comprehensively evaluating value orientation and understanding of large language models. *Proceedings of ACL 2024*; arXiv:2406.04214, 2024.
13. J. Yao, X. Yi, S. Duan, X. Wang, Y. Bai, M. Huang, P. Zhang, T. Lu, Z. Dou, M. Sun, and X. Xie. Value Compass Benchmarks: A platform for fundamental and validated evaluation of LLMs values. arXiv:2501.07071, 2025.
14. E. Coppolillo and E. Ferrara. MOSAIC: A multi-questionnaire benchmark of moral and social dispositions in LLMs. arXiv:2603.00048v2, 2026.
15. K. Zheng, Y. Ren, H. Yu, J. Leng, and D. Xiong. Beyond value benchmarks: Measuring value-structure alignment in large language models via symmetric Q-sorts. *Proceedings of ACL 2026 (Main)*, arXiv:2606.21939, 2026.
16. A. Kasat, A. Singh, A. Chadha, and A. Jain. Reasoning or rhetoric? An empirical analysis of moral reasoning explanations in large language models. arXiv:2603.21854, 2026.
17. J. Ji et al. MoralBench: Moral evaluation of LLMs. *ACM SIGKDD Explorations*, 27(1); arXiv:2406.04428, 2024.
18. K. Tanmay, A. Khandelwal, U. Agarwal, and M. Choudhury. Probing the moral development of large language models through Defining Issues Test. arXiv:2309.13356, 2023.
19. A. Khandelwal, U. Agarwal, K. Tanmay, and M. Choudhury. Do moral judgment and reasoning capability of LLMs change with language? A study using the multilingual Defining Issues Test. *EACL 2024 (Main)*, 2024.
20. Y. Chiu et al. MoReBench: Evaluating procedural and pluralistic moral reasoning in language models. *ICLR 2026*; arXiv:2510.16380, 2026.
21. D. Kiela et al. Dynabench: Rethinking benchmarking in NLP. *NAACL 2021*; arXiv:2104.14337, 2021.
22. Y. Zhou et al. RubricBench: Aligning model-generated rubrics with human standards. *Proceedings of ACL 2026 (Long)*, 2026.
23. J. Zhu and S. Lazar. Are LLMs bad at moral reasoning? arXiv:2606.11635, 2026.
24. J. Han, S. Song, H. Lee, and Y. Jo. Quantifying data contamination in psychometric evaluations of LLMs. *Findings of EACL 2026*; arXiv:2510.07175, 2026.
25. C. Spiesberger, D. Vazquez, N. Pochinkov, J. Gavenciak, T. Grietzer, G. Leech, and F. Schoots. Soft contamination means benchmarks test shallow generalization. arXiv:2602.12413, 2026.
26. Center for AI Safety and Scale AI. Humanity’s Last Exam. 2025.
27. C. White et al. LiveBench: A challenging, contamination-limited LLM benchmark. *ICLR 2025 (Spotlight)*; arXiv:2406.19314, 2025.
28. A. Al-Lawati, J. Lucas, D. Lee, and S. Wang. LLM benchmark datasets should be contamination-resistant. *ICML 2026 (Position Paper Track)*; arXiv:2605.19999, 2026.
29. M. Vaccaro. Preregistration for experiments with AI agents. *ICML 2026 (Spotlight, Position Paper Track)*; arXiv:2606.11217, 2026.
30. M. Thomas, K. Gligoric, and N. B. Shah. Mitigating LLM-based p-hacking by preregistering for the next LLM. arXiv:2606.27687, 2026.
31. A. Reuel, A. Hardy, C. Smith, M. Lamparth, M. Hardy, and M. J. Kochenderfer. BetterBench: Assessing AI benchmarks, uncovering issues, and establishing best practices. *NeurIPS 2024 Datasets and Benchmarks (Spotlight)*; arXiv:2411.12990, 2024.
32. A. Bean, R. Kearns, A. Romanou, F. Hafner, et al. Measuring what matters: Construct validity in large language model benchmarks. *NeurIPS 2025 Datasets and Benchmarks*; arXiv:2511.04703, 2025.
33. A. Ghosh, A. Reuel, J. Chim, K. Kennedy, S. Yadav, et al. Evaluation Cards: An interpretive layer for AI evaluation reporting. arXiv:2606.09809, 2026.
34. Y. Jiang, R. Zhang, K. Zhu, Y. Bai, S. Truong, X. Yi, S. Koyejo, and C. Xiao. AI evaluation should require standardized item-level data releases. arXiv:2604.03244, 2026.
35. P. Liang, R. Bommasani, T. Lee, et al. Holistic evaluation of language models. *TMLR*, 2023; arXiv:2211.09110.
36. S. Singh et al. The Leaderboard Illusion. *NeurIPS 2025 Datasets and Benchmarks Track*; arXiv:2504.20879, 2025.
37. K. Chehbouni, M. Haddou, J. Cheung, and G. Farnadi. Neither valid nor reliable? Investigating the use of LLMs as judges. arXiv:2508.18076, 2025 (preprint).
38. N. Calderon, R. Reichart, and R. Dror. The alternative annotator test for LLM-as-a-judge: How to statistically justify replacing human annotators with LLMs. *ACL 2025*, 16051-16081.
39. X. Yang, N. Feldhus, S. Mohtaj, L. Hennig, et al. What are we measuring in NLG? A meta-analysis of evaluation trends 2020-2025. arXiv:2601.07648, 2026 (preprint).
40. J. D. Norman, M. U. Rivera, and D. A. Hughes. Reliability without validity: A systematic, large-scale evaluation of LLM-as-a-judge models across agreement, consistency, and bias. arXiv:2606.19544, 2026.
41. H. Shen, N. Clark, and T. Mitra. Mind the value-action gap: Do LLMs act in alignment with their values? *EMNLP 2025 (Main)*; arXiv:2501.15463, 2025.
42. M. Wu, S. Sheng, W. Wang, Y. Yang, X. Sun, Y. Wang, J. Bu, and Y. Cao. The staircase of ethics: Probing LLM value priorities through multi-step induction. arXiv:2505.18154, 2025 (preprint).

43. H. Ye, J. Jin, Y. Xie, X. Zhang, and G. Song. Large language model psychometrics: A systematic review of evaluation, validation, and enhancement. *arXiv:2505.08245*, 2025.
44. A. R. Feinstein and D. V. Cicchetti. High agreement but low kappa: I. The problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6):543-549, 1990.
45. J. Ashkinaze, H. Shen, S. Avula, E. Gilbert, and C. Budak. The Deep Value Benchmark: Measuring whether models generalize deep values or shallow preferences. *NeurIPS 2025 (Spotlight)*; *arXiv:2511.02109*, 2025.
46. Y. Dong, X. Feng, C. Jiang, H. Ye, P. Zhang, and M. Song. Agent-ValueBench: A comprehensive benchmark for evaluating agent values. *arXiv:2605.10365*, 2026.
47. T. McCaslin, J. Alaga, N. Nedungadi, S. Donoughe, T. Reed, R. Bommasani, A. Painter, and L. Righetti. STREAM (ChemBio): A standard for transparently reporting evaluations in AI model reports. *arXiv:2508.09853*, 2025.
48. N. Chakraborty, S. Wang, and D. Jurgens. Structured moral reasoning in language models: A value-grounded evaluation framework. *arXiv:2506.14948*, 2025.
49. S. Jiao, S. Afroogh, K. Murali, D. Chen, D. Atkinson, and A. Dhurandhar. LLM ethics benchmark: A three-dimensional assessment system for evaluating moral reasoning in large language models. *Scientific Reports*, 15:34642, 2025.

582 **This appendix is part of the single submission PDF and inherits every page-1 label without ex-**
583 **ception:** PROVISIONAL · DESCRIPTIVE · EXPLORATORY · QUARANTINED PRE-GATE-M;
584 judges and coders UNVALIDATED and Gate M not run; Firewall G2 in both directions; profile sur-
585 faces NEVER-RANKED by construction; no ordering of any measured seat anywhere; exclusions
586 and re-labels rendered, never deleted; no new statistic computed.

587 **A Vocabulary, geometry, and the rendering law**

588 *Seat* — one model in one administration class. *Substrate* — seats administered together under one
589 route, rater stack and date; never merged. *Arc* — a profile-family administration class. *Stratum* —
590 one school by setting within one seat, or one administration class within a construct; the unit the size
591 labels (SMALL, FLOOR, CEILING, MODERATE) attach to. *Leg* — one rater's complete pass over
592 a substrate. *Judged* (competence) and *coded* (profile) are different operations on different families
593 and are never pooled. *Census-clean* — zero unexpected nulls against *scheduled* denominators, with
594 structural nulls pre-declared. *Firewall G2* — competence and profile results never blend, in either
595 direction.

596 The four competence settings are *S1a* fixed-key anchored items, mechanically scored and never
597 judged; *S1p* free production graded against a rubric; *S2* an agentic episode in a fictional organisation
598 with a mock tool world, where the rubric is scored on the post-episode debrief; and *S3* multi-turn
599 probing under pressure with mandatory bare-pressure and new-fact poles. The single profile setting
600 *P* is single-turn ecological elicitation coded into a profile code object.

601 **Per-family geometry — instrument extent, not results; the two families' counts are stated apart**
602 **and never compared.** The administered per-seat competence instrument is **120 subject cells** (*S1a*
603 *18 · S1p 12 · S2 48 · S3 42*), of which **102 are judged** (per school *S1p 4 · S2 16 · S3 14*). The
604 mandatory census phrasing is: *120 measured = 102 rubric-judged + 18 S1-anchored mechanical-*
605 *only items (no rubric by design)*. Item-side the same 120 is 40 per school. The 18 anchored items
606 carry no rubric reference, so the rubric-fidelity apparatus structurally cannot reach that stratum,
607 and the anchored setting is administered but unpublished for all 13 board seats, so no anchored
608 column appears on any board table. The per-seat profile frame is **274 cells** — 204 value-rating + 38
609 justification + 32 intuition — of which **70 are coded** (32 Moral Foundations + 38 Kohlberg), giving
610 pooled coded denominators $n = 512$ and $n = 608$, both endorsement-inclusive.

611 **Coding schemes and the sealed structure gate.** Each profile-side Moral Foundations cell is coded
612 four-way per foundation (raised, endorsed, argued against, absent) and rendered two ways side by
613 side — salient beside endorsed-only — never reduced to one. The sealed Stage-1 structure gate for
614 the Schwartz construct has three criteria on raw ratings: circular region-ordering correspondence at
615 least 0.6 of theoretical adjacencies preserved, an opposition sign pattern on two specified pairs, and
616 significance against a permutation null.

617 **Rendering law, in full.** Every surface carries its provisional label, rater identity, denominator and
618 stratum-size label before any observation is stated. Seats appear in census or source order, and the
619 order in force is named on the surface; **no ordering, tier, composite or best/worst statement of any**
620 **measured seat appears anywhere in this paper.** Coded legs are never collapsed, substrates are
621 never merged, and the two value-priority strata are never placed side by side. Excluded seats render
622 at denominator zero; a census flag never travels without its N/N count; dual-judged substrates get
623 one table per judge and the legs are never averaged; retractions, exclusions and re-labels are rendered
624 as labelled rows rather than deleted; and a superseded label is corrected in place with its correction
625 attached, never rewritten away.

626 **Cost of record, as process.** On the competence side four disjoint scopes exist and no permit-
627 ted source joins them; the four board substrates cost approximately \$5.69, carrying an unresolved
628 session-basis versus subjects-only incoherence as an open escalation label, with grading at \$0 cash.
629 On the profile side only a program cumulative lower bound of approximately \$65.27 is carried,
630 against a sealed halt-and-ask constant of \$450. Ceilings are not authorizations: every paid act re-
631 quires its own fresh probe, re-priced plan and explicit in-session approval, and a spent grant is
632 entered in a refusal register and refused by name.

633 **The blinding gate, in full.** A false-positive sweep over all public prompts is a build test. The banked
634 sign-off for the profile study records 0 hits, 0 disclosed and 0 blocking over 1,208 prompts, on a dry

run re-run against *real* subject records rather than fixtures. One genuine firing supplies the standing lesson: a path-fragment needle matched an ordinary English idiom in 2 of 1,208 prompts, and the adopted rule is never to soften a term, never to drop a prompt, and never to register a collision without an explicit ruling. A later coder run screened 560 prompts against 64 forbidden terms with 2 hits, 2 disclosed and 0 blocking, both hits being a vendor needle occurring inside an ordinary English word; the module runs its triage offline first, so the operator learns the verdict before the first call rather than on an abort partway through a shard.

B The competence coverage map and the board, by substrate

Competence surfaces only. Substrates never merged; census or source order, never an order of score. Standing banner applies.

substrate	seats	subject cells	judge leg(s)	OOF	verdicts/seat
B1	3	360/360	gpt-5.6-sol	single	102
B2	3	360/360	claude-opus-5dual		204
			.		
Frontier	2	240/240	gpt-5.6-sol		
			claude-opus-5single		102
			corrected leg		
B3	5	600/600	claude-opus-5dual		204
			.		
gpt-5.5	—	—	gpt-5.6-sol		
			EXCLUDED,	—	0
			reduced		
			roster		

*Table B1 — completion per substrate: B1 the Anthropic seats, B2 an open-weight trio, Frontier the OpenAI seats, B3 an open-weight extension. **This is a coverage map, not a board: no scores appear on it and no substrate is merged with another.** OOF is out-of-family status under the leave-one-out firewall, which is why judge stacks and verdict denominators differ by construction. On B1 the census flag reads false **solely** by way of the declared roster reduction, with an empty missing-records list and zero unexpected nulls over its 360/360 cells. The frontier substrate is census-clean at the subject* stage (240/240) while its corrected judge leg is 203/204 with one disclosed no-verdict — the only one on the board — leaving that seat’s Schwartz-agentic stratum at n = 15. Substrate cash lines were \$0, \$1.4782, \$0 and \$4.0052 on-tree respectively; on B2 the observed spend was 1.27 times a margined estimate of \$1.166 and the \$1.75 pause rail never fired.**

The seat index used by the board tables below. The index is a table-layout device with no ordering content of any kind; every seat slug is printed here in full.

index	seat	substrate
1	claude-opus-4-8	B1
2	claude-sonnet-4-6	B1
3	claude-haiku-4-5	B1
4	gemma-4-31b-it	B2
5	llama-4-scout	B2
6	glm-5.2	B2
7	gpt-5.4	Frontier
8	gpt-5.4-mini	Frontier
9	qwen3-8b	B3
10	mistral-small-3.2-24b	B3
11	kimi-k2.6	B3
12	deepseek-v4-flash	B3
13	qwen3-235b-thinking	B3

657 *Table B2 — the seat index. Indices 1-3, 4-6 and 7-8 follow census roster order within their substrate;*
658 *indices 9-13 follow the order the source edition prints the B3 seats, which is not the sealed roster*
659 *position order and not any score, tier or performance order. The index encodes nothing but row*
660 *position.*

661 The board readings follow at 189 board cells over 21 seat-rows. **Cells are per-seat mean scores**
662 **over that school-by-setting stratum on a 0-1 scale.** Column order is fixed: kohlberg S1 · S2 · S3,
663 then Moral Foundations S1 · S2 · S3, then Schwartz S1 · S2 · S3. **Pre-observation stratum-size**
664 **labels, set before any observation and identical on every board table below:** every S1 column
665 is SMALL at **n = 4** items per school per seat; every S2 column is MODERATE at **n = 16**; every
666 S3 column is MODERATE at **n = 14**. FLOOR means at or adjacent to 0.000 and CEILING at
667 or adjacent to 1.000; the ratified ceiling count on these legs is **ten single-turn Schwartz cells at**
668 **exactly 1.000.** Several cells sit in the FLOOR stratum and several S1 cells sit at CEILING; both
669 are disclosed design inputs and are **never read as seat verdicts.** Judges are UNVALIDATED; every
670 cell is a single administration; and the same 120 items reached the same 5 seats on two substrates,
671 so an independence-assumption caveat rides any cross-substrate statement.

672 **Table B3 — substrate B1 · judge gpt-5.6-so1, single out-of-family · 102 verdicts per seat ·**
673 **seats 1-3 in census roster order · PROVISIONAL, pre-Gate-M.**

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
1	0.890	0.979	0.874	1.000	0.966	0.682	1.000	0.817	0.386
2	0.890	0.920	0.873	0.980	0.972	0.822	0.500	0.938	0.891
3	0.672	0.804	0.808	0.980	0.960	0.662	0.750	0.909	0.643

674 *claude-opus-5 is in-family for these seats and is excluded by the leave-one-out firewall, so B1*
675 *carries no second leg and no agreement surface.*

676 **Table B4 — substrate B2 leg 1 · judge claude-opus-5, dual out-of-family · 102 judged cells**
677 **per seat · seats 4-6 in census roster order · PROVISIONAL, pre-Gate-M.**

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
4	0.390	0.857	0.494	0.955	0.846	0.177	1.000	0.861	0.516
5	0.585	0.673	0.391	0.485	0.386	0.206	0.919	0.867	0.464
6	0.835	0.963	0.972	0.955	0.988	0.684	0.750	0.964	0.686

678 **Table B5 — substrate B2 leg 2 · judge gpt-5.6-so1, dual out-of-family · 102 judged cells per**
679 **seat · seats 4-6 in census roster order · PROVISIONAL, pre-Gate-M.**

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
4	0.725	0.881	0.656	0.975	0.900	0.088	1.000	0.880	0.500
5	0.725	0.413	0.429	0.930	0.813	0.366	1.000	0.902	0.563
6	0.863	0.867	0.826	0.980	0.978	0.621	1.000	0.923	0.871

680 *Tables B4 and B5 — the two B2 legs, never averaged into a single number. A banked divergence*
681 *rides seat 5's Schwartz-S3 cell on leg 2: it is **0.563** in the edition of record and **.562** in an earlier*
682 *published readout — the same measurement at a different render of an exact decimal half, never*
683 *shown as equal and never averaged. A second banked provenance tension rides leg 1: no published*
684 *readout carries that leg's per-seat surface, which was produced by a post-hoc derived census, so its*
685 *primary source is the leg record tree via a verified recomputation rather than a readout table. Both*
686 *are carried, not resolved.*

687 **Table B6 — the Frontier substrate · judge claude-opus-5, single out-of-family, corrected leg**
688 **· 102 verdicts per seat · seats 7-8 in census roster order · PROVISIONAL, pre-Gate-M.**

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
7	0.918	0.434	0.900	0.955	0.463	0.516	1.000	0.565	0.582
8	0.863	0.295	0.838	0.955	0.355	0.401	1.000	[a] 0.466	0.505

Table B6 — [a] this stratum is over $n = 15$, not 16: one disclosed no-verdict, excluded and never imputed. A standing framing caveat rides any characterization of this substrate: the frontier band stays Claude-judged until ruled otherwise. **The predecessor of this leg is retracted and must never be quoted as a result:** on that leg the agentic debrief channel was empty for 48/48 cells on both seats against 0/48 on every other route for the identical items, so a blind judge faithfully scored an empty channel at approximately zero and a score-presence census signed off. The 96 agentic cells were re-administered (96/96 census-clean, 48/48 populated per seat) and the whole judge leg was re-run; the retracted-to-corrected movement by stratum was .000 to .434, .025 to .463, .050 to .565 on seat 7 and .000 to .295, .013 to .355, .062 to .466 on seat 8. Engagement was never the issue — tool use verified 16/16 and 14/16 on the spot-checked stratum; this is not a per-seat engagement count and no such figure exists.

Table B7 — substrate B3 leg 1 · judge claude-opus-5, dual out-of-family · 102 judged cells per seat · seats 9-13 in source-edition order · PROVISIONAL, pre-Gate-M.

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
9	0.000	0.688	0.151	0.480	0.738	0.000	0.500	0.673	0.257
10	0.223	0.643	0.225	0.895	0.778	0.000	0.725	0.759	0.346
11	0.640	0.930	0.806	0.730	0.889	0.476	0.750	0.944	0.473
12	0.184	0.933	0.731	0.938	0.799	0.226	1.000	0.947	0.329
13	0.613	0.684	0.425	0.725	0.625	0.133	0.719	0.709	0.089

Table B8 — substrate B3 leg 2 · judge gpt-5.6-sol, dual out-of-family · 102 judged cells per seat · seats 9-13 in source-edition order · PROVISIONAL, pre-Gate-M.

seat	koh S1	koh S2	koh S3	mft S1	mft S2	mft S3	sch S1	sch S2	sch S3
9	0.585	0.703	0.331	0.730	0.914	0.000	0.750	0.842	0.321
10	0.703	0.566	0.294	0.945	0.791	0.039	0.750	0.833	0.468
11	0.835	0.895	0.701	0.750	0.981	0.477	1.000	0.936	0.379
12	0.835	0.885	0.657	0.963	0.902	0.318	1.000	0.933	0.491
13	0.780	0.653	0.531	0.750	0.845	0.131	0.500	0.789	0.093

Tables B7 and B8 — seats 9-13 appear in the order the source edition prints them, which is **not** the sealed roster position order and **not** any score, tier or performance order. The two legs are never averaged.

Table B9 — substrate B3, the program's only banked grand means · leg 1 claude-opus-5, leg 2 gpt-5.6-sol · per-school columns at $n = 170$ each, leg mean over 510 cells, setting columns at $n = 60$ (free production), $n = 240$ (agentic), $n = 210$ (multi-turn) · PROVISIONAL, pre-Gate-M, both judges UNVALIDATED.

B3	kohlberg	mft	schwartz	leg mean	S1p	S2	S3
leg 1	.597	.518	.590	.568	.608	.783	.311
leg 2	.643	.594	.646	.628	.792	.831	.349

Table B9 — **the two legs are never averaged into one number.** Per-school and per-setting grand means for B1, B2 and the frontier substrate are **not published or banked anywhere**: they exist only as in-report recomputations and must never be called banked.

714 **Judge-pair agreement, on the two dual-judged substrates separately and never pooled.** Delta
715 is gpt-5.6-sol minus claude-opus-5 on the same cell; every figure is a property of the judge
716 pair under a prompt regime that has since been retired, so a repeat measures a different pair-regime.

statistic	B2 (306 paired cells)	B3 (510 paired cells)
mean absolute delta	0.1618	0.1476
median absolute delta	0.0500	0.0000
exact agreement	0.471	0.537
within 0.1 (epsilon convention)	0.670	0.749
within 0.1 (float-literal convention)	0.667	0.741
within 0.25	0.850	0.839
signed mean delta	+0.0403	+0.0599 [OPEN ESCALATION]
maximum absolute delta	1.00	1.00
final no-verdicts, either judge	0 (5 transient nulls re-graded)	0 on both legs

717 *Table B10 — PROVISIONAL · pre-Gate-M; both judges UNVALIDATED. Neither figure is validity.*
718 *Any side-by-side must fix one within-0.1 convention and name it; mixing the two published headlines*
719 *(.667 against .749) overstates the difference by the boundary cells alone (1 boundary cell on B2,*
720 *4 on B3). The signed-delta figure on B3 carries an open-escalation label wherever it is quoted.*
721 *Discordance concentrates in Schwartz multi-turn and kohlberg single-turn and multi-turn — a judge-*
722 *pair property, never a seat property.*

723 C Family-P surfaces: strata, agreement, and the endorsement containment 724 statement

725 **PROFILE ONLY from here — QUARANTINED-DESCRIPTIVE; NEVER-**
726 **RANKED by construction. Coders UNVALIDATED; Gate M pending; legs**
727 **never collapsed; substrates never merged; the two value-priority strata never**
728 **placed side by side.**

729 **Coded-surface completion, one row per substrate; never merged.**

substrate	seats	profile subject cells (274/seat)	coder legs (per leg, census-clean)
sealed core	5	1,970 / 1,970 / 1,926	875/875 records, 350 coded
extension 1	8	2,192 / 2,192 / 2,162	560/560, all coded
extension 2	3	822 / 822 / 813	210/210, all coded
out-of-pool	5	1,970 / 1,970 / 1,636	875/875 records, 350 coded
priority arm	5	9,280 / 9,280 / 9,231	0 legs (mechanical)

730 *Table C1 — QUARANTINED-DESCRIPTIVE; coders UNVALIDATED. Coverage is 16 of 16 seats*
731 *over the three pooled-eligible arcs (5 + 8 + 3); the four coded rows are seat-administrations, not dis-*
732 *tinct seats. The out-of-pool arc is cited for ordering replication only and is never pooled. Terminal-*
733 *unmeasured cells are never imputed and never dropped. The sealed core arc and the out-of-pool arc*
734 *each ran as one combined administration of 1,970 cells carrying 600 firewalled competence cells*
735 *per run; that 1,970 is **not** a profile denominator — the five-seat profile frame is 1,370. Coded-record*
736 *denominators by arc are 190 / 304 / 114 for kohlberg and 160 / 256 / 96 for Moral Foundations,*
737 *never summed across the fence. Coder-leg receipts record 0 no-verdicts on every leg of every arc,*
738 *with no leg aborted and none paused.*

739 **The two value-priority strata, in separated blocks, each carrying its own provenance label.**
740 **They are listed, never compared, never pooled, never differenced.**

741 *Block 1 of 2 — provenance: the regime of record, administered at the sealed*
742 *regime, 7 seats. 25,984/25,984 cells measured, complete, 0 re-run targets;*
743 *738 measured-invalid (718 missing-or-extra-rating, 20 duplicated-rating), dis-*
744 *closed rather than re-sampled because re-sampling would select for extractable*
745 *responses; a registered predicate divergence over exactly those 738 stands*
746 *banked and unreconciled; 4 genuine transport nulls recovered under an approved*
747 *wave. All 7 seats PERMITTED at the sealed 30-bundle floor; every seat ren-*
748 *ders “FAILED (inconclusive — power-limited)”, with signal-to-noise in the*
749 *band 0.605-0.887, all below the 1.5 anchor, so absence of structure may not*
750 *be claimed for any seat. Failed criteria across the arc are permutation-p 7,*
751 *opposition-sign 4, circular-order 3. At observed signal-to-noise, 30 bundles per*
752 *seat is under-powered per the sealed simulation — power 0.89 at signal-to-noise*
753 *at or above 1.5, 0.36 at approximately 1.0, at n = 30 with 0 false positives on noise*
754 *— a property of the design and the condition, never of a seat. A Rician up-bias*
755 *caveat rides every value in this block.*

756 *Block 2 of 2 — provenance: a disclosed stratum at the MEASURED regime (tem-*
757 *perature 0), which is not the regime this arm was labelled with; 5 seats. 9,280*
758 *scheduled and 9,280 measured, 0 judge invocations, 0 re-run targets; strict extrac-*
759 *tion succeeded on 8,652 of 8,700 bundle-input records (0.9945) with 48 honestly*
760 *unscored and never imputed (0.0055), and 0 recovered by the sealed path. All five*
761 *seats reached 30 bundles at the floor and PERMITTED, with 0 rows dropped from*
762 *the structure math; the arm tally is PASSED 1 / FAILED 4, observed signal-to-*
763 *noise 0.523-0.898 — and recovery is not validity: a PASSED recovery says the*
764 *structure gate was met, never that the reading is valid. Per-seat priority matrices*
765 *and Stage-2 vectors for this stratum live in the machine record only and are never*
766 *rendered. A Rician up-bias caveat rides every value in this block as well.*

767 **The four seats neither stratum reaches.** The four provider-impossible seats fall in **two distinct**
768 **error classes** and had **no cell administered**, so by construction no per-seat Schwartz figure exists
769 for any of them. All four had passed a reachability preflight at a *different* parameter, which is the
770 lesson the standing probe rule encodes: a preflight cannot see an impossibility at a parameter it does
771 not send.

772 **Coded-channel agreement, by arc, never pooled; column order is pooled / confirmatory / ext1**
773 **/ ext2.** Agreement is not accuracy, and no channel-reliability statement of any kind is available for
774 the Schwartz channel, which is absent by construction.

kohlberg field	n = 608	n = 190	n = 304	n = 114
register	0.933	0.884	0.977	0.895
primary	0.827	0.847	0.816	0.825
secondary	0.648	0.653	0.651	0.632
confidence	0.538	0.505	0.595	0.439

foundation	n = 512	n = 160	n = 256	n = 96
care	0.785	0.825	0.754	0.802
equality	0.697	0.688	0.680	0.760
proportionality	0.738	0.769	0.734	0.698
loyalty	0.660	0.650	0.660	0.677
authority	0.689	0.631	0.703	0.750
purity	0.943	0.938	0.945	0.948

775 *Tables C2 and C3 — QUARANTINED-DESCRIPTIVE; coders UNVALIDATED. register abbreviates the register_without_structure field. Kohlberg fields are in the sealed coder-schema*
776

777 field order; and foundations in sealed instrument order; neither is sorted by a numeric column, and
778 foundation shares are **never averaged into one number**. The confidence channel is reported-but-
779 non-analytic by amendment: it is always reported with its own denominator and never used to
780 support a claim, enter a composite, condition an inclusion rule or gate any other figure. On the
781 out-of-pool arc the same kohlberg field order gives 0.947 / 0.916 / 0.737 / 0.579 at n = 190 — the
782 **ordering** replicates; the numbers are never pooled. One seat-adjacent figure exists and it names no
783 seat: the confidence band falls to 0.421-0.447 on the three coverage-completion seats. No per-seat
784 coder-agreement figure exists anywhere.

Cohen's kappa [7]	confirmatory	ext1	ext2
care	0.483	0.485	0.564
equality	0.507	0.508	0.591
proportionality	0.655	0.609	0.544
loyalty	0.497	0.530	0.544
authority	0.481	0.559	0.635
purity	0.772	0.795	0.816
kohlberg primary	0.716	0.658	0.698

785 *Table C4 — QUARANTINED-DESCRIPTIVE. Kappa is never pooled: there is no pooled kappa*
786 *column and none may be computed. Kappa is emitted for these seven fields only; the secondary,*
787 *register and confidence channels have no chance-corrected counterpart anywhere in the banked*
788 *artifacts. The out-of-pool arc's kohlberg primary kappa is 0.848 and is never pooled with the three*
789 *above. Purity's comparatively high figures are partly a base-rate effect — it carries the smallest*
790 *salient share on every seat, mostly 0.094-0.219 — but not only* one [44].**

791 **The validator battery, at class grain.** Construct-coverage quotas require every Moral Foundations
792 foundation as a primary target at least three times, all four Schwartz higher-order quadrants at least
793 four times, and all three neo-Kohlbergian schemas as production targets at least four times each,
794 with Liberty admitted only as unscored context. A contamination screen forbids 8-gram overlap
795 with canonical instrument strings. A scorer round-trip gate refuses to admit an item until its scorer
796 has round-tripped synthetic passing, failing and malformed responses, returning non-null in-range
797 for the first two and a clean unscored — not a crash, not a zero — for the third. A profile-variant
798 integrity gate requires every profile item's variant group to hold at least three members with the same
799 construct targets, non-overlapping surfaces and a social-desirability direction annotation. The twin-
800 integrity gate is §3.4's. Every profile item carries an annotated socially-most-desirable response
801 direction argued per item, so a social-desirability null profile is constructible **without administering**
802 **any such scale verbatim** — an annotation present on all 142 items and, as disclosed, consumed by
803 no analysis code.

804 **The stated-endorsement containment statement.** A sealed cross-cutting layer of 24 items — 8
805 per school — carries a stated-endorsement tag. A twin-fidelity review covered **16 of those 24** (two
806 schools) and found **8 of the 16 reviewed pairs confirmed defective (50.0%, Clopper-Pearson [8]**
807 **95% CI 24.7-75.3%)**, all in one class; those 8 are excluded from any stated-endorsement analysis,
808 claim or surface, with twins excluded identically. Three denominators are distinct and never inter-
809 changed: 16 is the review's universe, 24 is the layer extent, and 512 and 608 are the pooled coded de-
810 nominators, which are **endorsement-inclusive**. The containment statement is printed rather than the
811 denominator revised: the 6 excluded intuition-frame endorsement identifiers contribute **96 of the 512** cells from
812 inside the denominator, the 8 justification-frame endorsement identifiers contribute **128 of the 608**,
813 and the 2 excluded value-rating identifiers enter neither figure because that channel is mechanically
814 extracted and reaches no coder. **No denominator is revised, no agreement rate is restated, and**
815 **no exclusion-adjusted figure exists anywhere** — one would be a new statistic, and this paper com-
816 putes none. The exclusion is **directional**, having removed one school's purity endorsement probe
817 and both of another school's power-pole items, with the consequence that the sealed six-foundation
818 simplex is not evaluable on the surviving layer: purity is coded non-absent in **0 of the 64** surviving
819 stated coder cells, and no number for it is computed. The third school's eight pairs were later re-
820 viewed and found **3 of 8 confirmed defective (37.5%, CI 8.5-75.5%)**, all in the domain/stakes class,
821 with **0 of 8** in the class that carried all eight defects in the sibling arms; the intervals overlap heavily
822 and **no cross-arm rate claim is made**. Those reviewers are themselves unvalidated. Two standing
823 bars follow: no scored consumption of that coding channel until an administered path consumes its

824 repaired rubric, and no per-cell public-versus-twin comparison on the **76 of 142** single-variant profile
825 items until an administered-letter order-synchronization design exists, since the picker draws each
826 half's administered letter independently and 5 of 8 reviewed pairs were administered at mismatched
827 letters.

828 D Instrument extent and disclosure debt

instrument fact	value
public items (sealed)	422 — competence 280 · profile 142
competence items by school	mft 84 · schwartz 116 · kohlberg 80
profile items by school	mft 32 · schwartz 72 · kohlberg 38
distinct rubrics	368 — mft 110 · schwartz 182 · kohlberg 76
competence items carrying a shadow identifier	280 / 280 — the only shadow count carried anywhere; no instrument-wide shadow-twin total is asserted
of the competence 280	120 board-administered (42.9%) · 160 NEVER administered, administration BLOCKED, 0/160 measured
the 160, by wave	144 + 16 = 160
items with a twin baseline at temperature zero	406 / 422
items with a recorded exposure	406 / 422; 16 shadow surfaces at zero exposure — a pre-declared release precondition is not met instrument-wide
instrument-wide vintage split	pre-expansion 262 · first expansion wave 144 · later wave 16
profile items carrying a variant group	142 / 142, in 142 distinct groups; zero competence items carry the field

829 *Table D1 — instrument extent and its standing disclosures. Item counts describe extent, not results;*
830 *under Firewall G2 no result is joined across families. The 160 expansion items are pool extent only*
831 *and are **never** implied to be measured. Three distinct objects numbered 262 exist in this program —*
832 *262 judged cells per seat in an unreachable target frame, 262 pre-expansion public items, and 262*
833 *twin pairs — and they are never interchanged.*

834 Two disclosures bind every citation of these figures. First, the expansion administration is blocked
835 and its registry field is false for all 160, with the identifier-set intersection against every competence
836 run tree empty; a pool-wide pass on the open defect class is recommended before administration is
837 defensible, and no pool-wide rate for that class exists. Second, the profile pool has been unchanged
838 at 142 throughout the entire expansion arc — all 160 additions landed in the competence family —
839 so no profile figure moves under any expansion statement.

840 **The fidelity census, in strata.** The pre-registered **n = 18** sample that returned STOP commissioned
841 a full census, which found **102 of 144 expansion pairs (70.8%, Clopper-Pearson 95% CI 62.7-**
842 **78.1%)** carrying at least one confirmed defect. The repair amendment then took rubrics defective
843 under the standard threshold from **110/144 to 0/144**, twins byte-untouched, moving no measured
844 board value because none of the 144 has ever been administered. The census is census-grade —
845 all 144 checked, not sampled — and its restricted 126-pair rate of **71.4%** (CI 62.7-79.1%) was
846 statistically indistinguishable from the carried **n = 18** estimate of **66.7%** (CI 41.0-86.7%). The
847 dominant class is a rubric-echo class at **94 of 144 = 65.3%** (CI 56.9-73.0%); per-class confirmed
848 defects run 18 / 0 / 94 / 0 / 2 / 3 across six classes with 13 / 0 / 4 / 0 / 3 / 5 escalated as unclear. By
849 school, defect shares are 34 of 40 (85.0%, CI 70.2-94.3) for Moral Foundations, 42 of 64 (65.6%, CI
850 52.7-77.1) for Schwartz and 26 of 40 (65.0%, CI 48.3-79.4) for kohlberg; by setting, 4 of 8 (50.0%,
851 CI 15.7-84.3) free-production, 51 of 67 (76.1%, CI 64.1-85.7) agentic and 47 of 69 (68.1%, CI 55.8-
852 78.8) multi-turn. Class counts are defect *cells*, not mutually exclusive, and do not sum against the
853 102 any-class defect *pairs*. **An honesty note travels with the repair:** 8 additional cells were fixed
854 in-session from the spot-check and **were not re-reviewed blind.**

855 **The two twin-baseline epochs, per epoch and never pooled.** The first epoch administered 262 twin
856 pairs by 2 halves by 2 subjects (1,048 cells) at $n = 120$ competence pairs per model; under a two-one-
857 sided-tests procedure at a 0.10 margin with an item-pair-clustered bootstrap display interval (20,000
858 draws, seed 0) both seats returned **EQUIVALENT** — gaps **+0.0296** and **+0.0304**. Its one flagged
859 stratum, kohlberg on one seat at **+0.1475** with a display interval excluding zero, does not reproduce
860 at the stronger operating point and is **an authoring-asymmetry signal, never a contamination**
861 **claim**. The second epoch administered 144 pairs by 2 halves by 2 seats (576 cells) at $n = 144$ pairs
862 per model: one seat **+0.0321** (sd 0.371, $p = 0.0149$, **EQUIVALENT**), the other **+0.0603** (sd 0.398,
863 $p = 0.117$, **NOT demonstrated equivalent**), the latter concentrating in kohlberg at **+0.104** while
864 the first seat's kohlberg gap runs the opposite direction at **-0.045**. These are temperature-zero pre-
865 exposure twin-parity (authoring) signals, **not** contamination results; the epochs are never pooled,
866 averaged or differenced, and the attenuation direction above rides on all of them. A third, wave-D
867 epoch was never run and contributes no result.

868 E Governance chronology, at header grain

869 The amendment register's headings run **A-1 through A-22**. A change to a sealed constant, pool or
870 rubric requires a ledger entry recording old value, new value and reason; entries that only read sealed
871 state, or that record a status, rotate nothing, and the ledger records that distinction rather than im-
872 plying a uniform rule. At header grain: A-1 scoped the pipeline order to the confirmatory track and
873 created the labelled provisional-board carve-out; A-2 re-scoped a rubric-tier count on the agentic
874 board; **A-3 never existed as a ratified amendment** — its draft was withdrawn as unnecessary after
875 a mandatory probe found a judge alive that a carried belief had declared dead, and the heading is
876 retained rather than renumbered; A-4 extended the Gate-M substrate; A-5 through A-9 are item- and
877 twin-level repairs and annotations (a shadow system-prompt re-authoring, per-variant rating keys for
878 seven order-divergent items, two residual-echo apparatus fixes, a symmetric distractor repair, and a
879 variant-to-rating-order annotation on 22 public items and their 20 twins, all 60 of 60 identical to the
880 public twins); A-10 scoped an administration temperature for one arm only; A-11 added an adminis-
881 tration dimension to the bundle key; A-12 was a combined expansion-repair event with one re-seal;
882 A-13 sealed the evaluation rules for two structure-recovery criteria, the power-limited reporting rule,
883 and the temperature-zero scope closure; A-14 was the expansion rubric repair, executed under the
884 seal invocation of record; A-15 sealed the value-priority administration regime and a substrate-join
885 rule; A-16 repaired the judge template's conformance to its binding specification in both builders as
886 one event; A-17 replaced the A-15 join with the ruled two-stratum disclosure; A-18 pinned a read
887 path to the pre-expansion vintage; A-19 admitted the five re-administered seats per census-clean seat
888 and retired the temperature-zero stratum per outcome; A-20 pinned three analysis functionals **blind**,
889 attested by the fact that no executable path existed to compute them and no data look occurred; A-
890 21 combined a twin re-authoring with a coding-rubric repair in one event, under its own licensed
891 seal invocation; and A-22 recorded one seat's reduced-roster status at amendment grain, editing no
892 sealed constant, rotating no hash and engaging no seal.

893 Three properties of this ledger are load-bearing. **Annotate, never renumber**: a superseded figure is
894 corrected in place with its correction attached, and an appended annotation that moves a file digest
895 discloses the movement rather than hiding it. **A read-only amendment is still an amendment**:
896 two entries rotate nothing, because they read sealed constants and never write them, and the ledger
897 records that distinction explicitly. And **posture is recorded before administration**: the per-seat ad-
898 mission rule of A-19 was fixed before any of its seats ran, which is what keeps the stratum semantics
899 result-blind rather than a post-hoc choice per seat.

900 F The nearest systems, side by side

901 Every cell below is taken from a verified row of our related-work sweep and its blind-verification
902 annex; a dash means the property is **not established by that row**, not that the system lacks it.
903 Nothing in these tables is a quality comparison, and no row is ordered against another.

system	MFT	Schwartz	Kohlberg	theory-native construct	judge human-validated
ValueBench [12]	—	—	—	no	—
Value Compass [13]	—	yes	—	yes	—
MOSAIC [14]	yes	yes	no	no	—
Q-sort structure work [15]	yes	yes	no	yes	—
Kohlberg free-text scoring [16]	—	—	yes	yes	—
MoralBench [17]	yes	no	no	—	no
MoReBench [20]	no	no	no	—	yes
this work	yes	yes	yes	yes	no

904 *Table F1 — school coverage and judge status. “Theory-native construct” means the system scores*
905 *each school in that school’s own measurement vocabulary rather than a shared scale. **Our own***
906 *judge-validated cell reads “no”, and that is the honest state of this work: the expert gate is designed*
907 *and has not run.*

system	competence/disposition		per-item twins	preregistered governance
	split reported	profile ranked		
ValueBench [12]	yes	—	—	—
Value Compass [13]	yes	yes	—	—
MOSAIC [14]	no	—	—	—
Q-sort structure work [15]	no	—	—	—
Kohlberg free-text scoring [16]	no	—	—	—
MoralBench [17]	no	—	—	—
MoReBench [20]	no	—	—	—
this work	yes	no, by construction	yes	yes

908 *Table F2 — reporting and governance properties. The only two rows that report both families*
909 *separately are ValueBench and Value Compass; of those, Value Compass ranks its conformity fam-*
910 *ily, which is the exact contrast the firewall of §3.2 draws. “Per-item twins” means a paired pub-*
911 *lic/shadow twin per item with an exposure ledger, not a withheld sample. Item counts and design*
912 *properties describe extent, not results; under Firewall G2 no result is joined across families.*

913 G An illustrative competence item, at class grain

914 **Illustrative example authored for this paper; not a pool item. No pool item, shadow twin,**
915 **rubric text or model response appears here or anywhere in this paper.** It exists to show the
916 *shape* of a competence item and of what the judge returns, at the grain the custody law permits.

917 **Scenario sketch (invented for this illustration).** A municipal transit office must decide whether
918 to keep a late-night bus route that serves a small number of shift workers at high cost per rider, or
919 redirect the money to a daytime route serving many more people. A long-serving dispatcher has

920 publicly promised the shift workers the route would stay. The model is asked to reason about the
921 decision *within a named school*, and — this is the anti-cueing gate in action — the item surface
922 never names the school, its foundations, its values or its schemas. Only the rubric and the coding
923 instructions speak theory.

924 **What the rubric gates, at class grain.** The rubric does not reward a verdict; it gates the *structure*
925 of the reasoning. Its criterion classes ask whether the response detects the considerations the named
926 school treats as morally live in this scenario; whether it distinguishes them from adjacent considera-
927 tions a caricature of that school would conflate them with; whether it holds the trade-off open rather
928 than collapsing it to a slogan; and whether it sustains that structure under the adversity the setting
929 adds. Two guards ride every such rubric: an adequacy-scale guard forbids comparatives that would
930 rank one school's structures against another's, and a political-valence balance requirement applies
931 to items keyed to one designated schema class.

932 **What a judge verdict object contains, structurally.** Per criterion, a label rather than a bare num-
933 ber; a gate outcome where a gate applies; and — under the repaired template epoch — a verbatim
934 quoted span from the response supporting the label, with a payload missing an owed quote rejected
935 as *unscored* rather than scored zero. The per-cell object also carries the rater identity, the cell's
936 school-by-setting stratum, and the score field the census reads. **Every grade reported in this paper**
937 **predates that repair**, which is exactly why §8 states that no landed grade can be checked against
938 the text that produced it.